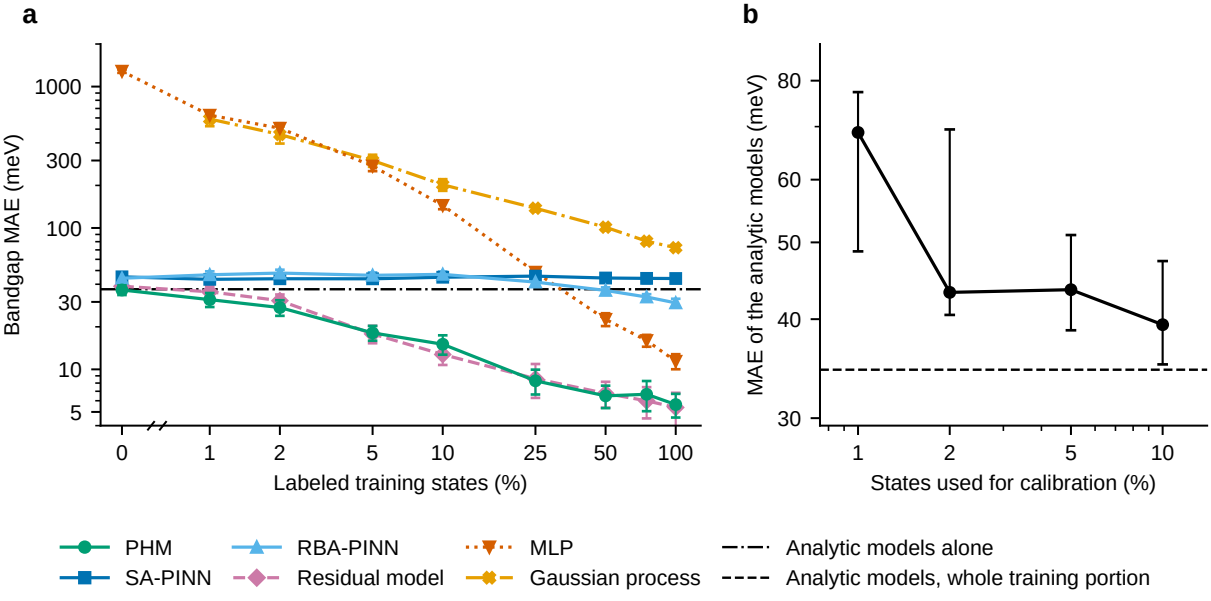
a
b
Bandgap MAE (meV)
1000
300
100
30
10
5
0
1
2
5
10
25
50
100
Labeled training states (%)
MAE of the analytic models (meV)
80
60
50
40
30
1
2
5
10
States used for calibration (%)
PHM
SA-PINN
RBA-PINN
Residual model
MLP
Gaussian process
Analytic models alone
Analytic models, whole training portion